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RESEARCH HIGHLIGHT

Vascular gatekeepers of central nervous system immunity

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INTRODUCTION

Neuroimmunology has traditionally focused on immune processes within the brain parenchyma, where responses in central nervous system (CNS) disease are interpreted through blood–brain barrier dynamics, leukocyte infiltration from the circulation, and glial activation. Over the past decade, however, vascular networks at CNS borders have emerged as important sites of immune regulation and fluid exchange.[1]

The identification of distinct anatomical routes linking the brain, meninges, nasal regions, and skull bone marrow (SBM) has reframed CNS borders as an immunovascular interface organized around cerebrospinal fluid (CSF) flow.[1] Within the brain, interstitial fluid and CSF can exchange via perivascular glymphatic pathways, enabling the clearance of metabolites and antigens into the CSF. Rather than remaining compartmentalized, this CSF is routed to multiple exit sites, including arachnoid cuff exit (ACE) points at bridging veins, spinal nerve sheaths, and the cribriform plate, where it gains access to border tissues. Across these sites, CSF-derived signals are sampled by immune cells and drained via meningeal and nasal lymphatic vessels to cervical lymph nodes, establishing these pathways as key sites for immune surveillance. In parallel, skull–meningeal channels surrounding vascular connections provide direct conduits between the dura and SBM, linking CSF-exposed compartments with local bone marrow reservoirs (Figure 1).

Crucially, these pathways are not unidirectional routes but sites of active exchange.[1] Skull–meningeal channels enable SBM-derived immune cells to traffic into the dura, independently of blood circulation, where they encounter CSF-derived signals and shape local responses. Similarly, immune activity at these borders can feed back to the brain via CSF to influence neuronal function.

Together, these features define CNS borders as sites where vascular and perivascular networks coordinate fluid dynamics and immune trafficking. Building on this framework, we highlight recent advances that describe how blood and lymphatic vasculature of this interface actively regulate immune function, and how its modulation offers therapeutic strategies in CNS disease.

Keywords
lymphatics, meninges, neuroimmunology, skull bone marrow

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MENINGEAL BLOOD VESSELS REGULATE IMMUNE SURVEILLANCE AT CNS BORDERS

The meninges contain dense vascular networks that provide key routes for immune cell entry at CNS borders. While prior works described how leptomeningeal and dural vasculature enables immune cell extravasation,[2] Hohsfield *et al.* identified the velum interpositum as a previously unappreciated meningeal route for immune trafficking.[3] This leptomeningeal extension, embedded beneath the hippocampus, enables peripheral myeloid populations to access the CNS during inflammation, as shown by fate mapping and intravital imaging.

Extending outwards, cerebral bridging veins drain into the dural sinuses within the dura mater. While the arachnoid barrier separates the CSF-filled subarachnoid space from the dura, this barrier is interrupted at sites of bridging vein passage. Here, ACE points permit CSF egress into the dura allowing immune surveillance and waste clearance at CNS borders.[4] Recent work shows that they are not simply passive conduits but are actively regulated. Mamuladze *et al.* found that mast cells are strategically positioned at ACE points and control bridging vein dilation through histamine release, thereby altering perivascular

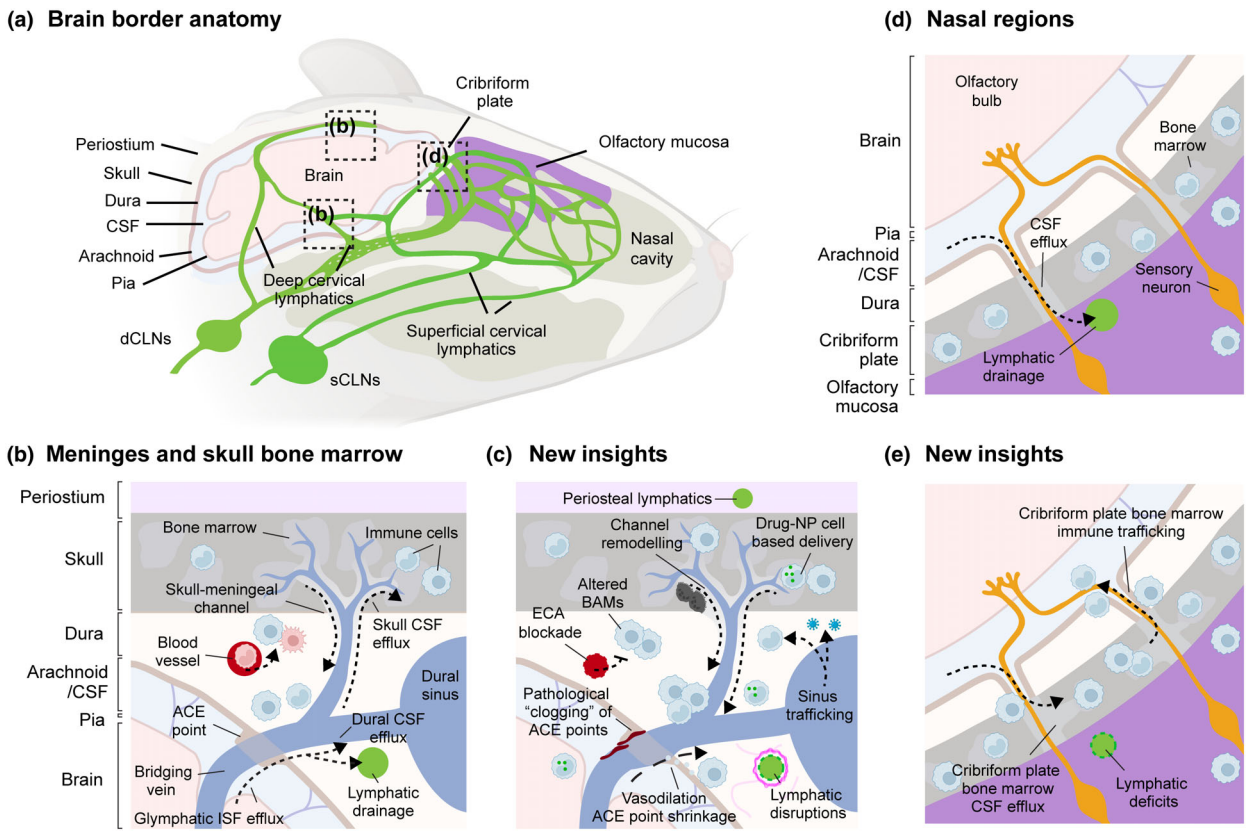

Figure 1. Vascular and perivascular connections between the skull, meninges, olfactory mucosa, and central nervous system (CNS) instruct brain border immunity and offer novel therapeutic targets. **(a)** Schematic representation of the brain borders including the cerebrospinal fluid (CSF), lymphatic networks, meninges (dura, arachnoid, and pia), skull, and cribriform plate at nasal regions. **(b)** Enlarged schematic of the meninges and skull bone marrow (SBM) highlighting how perivascular and vascular pathways at arachnoid cuff exit (ACE) points, SBM-meningeal channels, and blood and lymphatic vessels enable CSF flow and immune cell trafficking. **(c)** New insights highlight how ACE points can be modulated through mast cell-driven vasodilation or pathological protein deposition which restricts CSF access to the dura mater and trafficking of immune cells or infectious pathogens to the CSF. Immune cell access to the dura can be modulated by blockade of key vessels, by adjusting sinus tone, or through osteoclast bone remodeling of osseous channels. SBM also undergoes remodeling in disease and provides a novel route to deliver drugs to the brain. Lymphatic drainage from the dura can be impaired by increased collagen deposition and tissue stiffening. **(d)** Enlarged schematic of nasal regions highlighting how sensory neuronal projections through the cribriform plate enable CSF to reach the olfactory mucosa and be drained by nasal lymphatics. **(e)** New insights highlight how bone marrow channels are also present in the bone of the cribriform plate, and following inflammatory insults, resident immune cells here can migrate via osseous channels adjacent to olfactory sensory neurons and participate in immune responses. BAMs, border-associated macrophages; dCLNs, deep cervical lymph nodes; ECA, external carotid artery; ISF, interstitial fluid; NP, nanoparticle; sCLNs, superficial cervical lymph nodes.

space size and restricting CSF access to the dura.[5] In bacterial meningitis, mast cell activation and ACE point shrinkage prevent pathogen dissemination to the subarachnoid space, protecting the CNS from infection.[5] Similarly, Smyth *et al.* showed that in Alzheimer's disease models, amyloid-β deposition around bridging veins disrupts vessel tone and obstructs perivascular outflow, restricting CSF flow through these exit pathways.[6] These studies highlight how immune signaling and pathology shape CSF and pathogen exchange between CNS border compartments.

ACE points also provide a route through which dural immune populations can influence the brain by enabling the passage of secreted factors. In this context, Marin-Rodero *et al.* identified a population of meningeal regulatory T cells that produce immunomodulatory factors capable of maintaining neural homeostasis and limiting neuroinflammation, highlighting how immune populations at CNS borders can exert control over brain physiology.[7]

Within the dura, additional vascular compartments orchestrate neuroimmune functions. Gao *et al.* demonstrated that dural blood vessels regulate antitumor immunity

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in glioblastoma, showing that obstruction of the external carotid artery, which supplies blood to the dura, enhances the accumulation of resident border-associated macrophages, promoting meningeal T-cell activation and tumor rejection.[8] At a larger scale, Monaghan *et al.* found that the dural venous sinuses undergo dynamic alterations in vascular tone by smooth muscle cells, more reminiscent of arteries than veins.[9] By modulating sinus endothelial signaling, they demonstrated how these endothelial cells dynamically adjust their cell-to-cell junctions thereby altering leukocyte adhesion and pathogen entrance at CNS borders.[9]

Together, these studies demonstrate that meningeal blood vessels are active regulators of CNS border function, where vascular structure and tone govern fluid dynamics, immune cell trafficking, and local immune defense.

SKULL BONE MARROW PROVIDES A LOCAL RESERVOIR FOR IMMUNE RESPONSES TO CNS PATHOLOGY

Beyond meningeal blood vasculature, further advances have refined our understanding of SBM-vascular connections. Eme-Scolan *et al.* demonstrated that channels connecting the SBM to the dura undergo remodeling during viral neuroinflammation, driven by osteoclast-mediated restructuring of bone architecture.[10] This remodeling alters the diameter and permeability of these perivascular channels, enhancing immune cell trafficking into the dura. Similarly, Laaker *et al.* showed that analogous bone marrow channels exist within the cribriform plate, enabling bone marrow-derived immune cells to traffic adjacent to olfactory nerves and contribute to neuroimmune activation.[11] These findings extend the concept of bone marrow–to–meningeal immune communication beyond the calvarium to compartments in the nasal regions.

The relevance of SBM–CNS connections in humans has also begun to emerge. Dubey *et al.* demonstrated that glioblastomas induce remodeling of overlying SBM, altering the immune landscape of the skull and potentially providing a reservoir for tumor-associated immune infiltration.[12]

These findings suggest that SBM connections represent a dynamic immunological compartment that can be reshaped during pathological conditions.

LYMPHATIC DRAINAGE CONTROLS CSF CLEARANCE AND NEUROIMMUNE COMMUNICATION

Similar to the new insights into blood vasculature, additional work has reshaped our understanding of CNS lymphatic pathways. Jin *et al.* explored CSF outflow routes and demonstrated that lymphatic vessels surrounding the pterygopalatine arteries and continuing from nasal regions as superficial cervical lymphatics represent additional drainage pathways for CSF to superficial cervical lymph nodes.[13] Papadopoulos *et al.* further showed that different lymphatic outflow pathways contribute unequally to CSF homeostasis, and that physiological variables such as lymphatic pumping, body positioning, and intracranial pressure differentially influence the contributions of these nasal and meningeal lymphatic CSF drainage routes.[14]

Adding further anatomical complexity, Chang *et al.* demonstrated that lymphatic vessels are present within the skull periosteum, a layer of connective tissue overlying the skull, but absent from the bone itself.[15] Although the relevance of periosteal lymphatics to CNS immunity remains unclear, this suggests further potential routes for immune and fluid exchange at the CNS borders.

Advances in our understanding of what drives CNS lymphatic dysfunction have also been made, which may yield new therapeutic targets. Jin *et al.* identified mechanisms contributing to age-related decline in lymphatic drainage, including reductions in lymphatic valves and altered response to nitric oxide signaling.[13] Similarly, Hitpass Romero *et al.* showed that extracellular matrix remodeling surrounding aged meningeal lymphatic vessels impairs their drainage capacity.[16]

BRAIN-BORDER VASCULAR PATHWAYS PROVIDE OPPORTUNITIES FOR THERAPEUTIC INTERVENTION

Building on an improved understanding of CNS border vasculature, recent studies have begun to explore how these sites can be therapeutically manipulated. Jin *et al.* demonstrated that noninvasive mechanical stimulation of superficial cervical lymphatics enhances CSF drainage and brain waste clearance, highlighting how peripheral interventions can influence CNS homeostasis.[13] In parallel, Ma *et al.* investigated lymphaticovenular anastomosis, a surgical procedure connecting CNS-draining lymphatic vessels to the venous circulation, as a strategy to augment lymphatic drainage in Alzheimer's disease patients.[17] However, the long-term efficacy and safety of this procedure remain uncertain. Clinical relevance

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of these drainage pathways was supported by Al-Diwani *et al.*, who reported that CSF-derived neurodegeneration biomarkers accumulate within human cervical lymph nodes, indicating that these structures serve as downstream reservoirs of CNS-derived signals.[18]

Beyond lymphatic modulation, other studies explored how CNS border vasculature may be harnessed for therapy. Gao *et al.* demonstrated a clinically safe and feasible approach in which SBM–derived immune cells can be exploited as endogenous carriers for CNS drug delivery.[19] Following intracalvarial injections of drug-nanoparticle conjugates, they demonstrated how SBM immune cells can incorporate these and use SBM channels to traffic to the brain where they deliver their therapeutic cargo to improve outcomes in stroke models.[19]

FUTURE DIRECTIONS AND PERSPECTIVES

These studies converge on the concept that CNS borders are dynamic, actively regulated interfaces, where vascular architecture controls the movement of fluids, immune cells, and signals between the brain and the periphery, shaping neuroimmune communication.

These insights also highlight CNS borders as attractive therapeutic targets. Because dural vasculature, SBM, and lymphatics reside outside of the BBB, they represent accessible sites for modulating immune responses, enhancing clearance of pathological proteins, or delivering therapeutic agents to the brain. Future strategies that exploit these pathways to modulate border vascular function, recruit beneficial immune populations, or harness SBM trafficking may offer opportunities to influence CNS disease.

Ultimately, these advances position the vascular systems of the CNS borders as central regulators of brain immunity and promising targets for the next generation of neuroimmunological therapies.

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AUTHOR CONTRIBUTIONS

Luca Vinnell: Conceptualization; writing – original draft. **Kate Hitpass Romero:** Writing – review and editing. **Taylor J Stevenson:** Writing – review and editing. **Justin Rustenhoven:** Conceptualization; writing – original draft.

CONFLICT OF INTEREST

The authors declare no conflicts of interest.

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